

CMI DS Collegiate Programming Contest 2026

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Problem A: Avoid the Collisions

You are given a sequence of n distinct positive integers:

$$B[1], B[2], \dots, B[n],$$

and a sequence of integer labels:

$$G[1], G[2], \dots, G[n],$$

where $G[j]$ is the label assigned to index j .

A subset of indices $S = \{j_1, j_2, \dots, j_m\}$ is called **Valid** if for every pair of distinct indices $j_p \neq j_q$ in S :

$$G[j_p] \neq G[j_q].$$

That is, no two chosen indices share the same label.

The **Weight** of a Valid subset S is defined as:

$$W(S) = \sum_{j \in S} B[j].$$

Your task is to find the maximum possible Weight over all Valid subsets of size at most M .

Input

- The first line contains two integers n and M .
- The second line contains n distinct positive integers $B[1], B[2], \dots, B[n]$.
- The third line contains n integers $G[1], G[2], \dots, G[n]$, the label of each index.

Output

Print a single integer, the maximum Weight of a Valid subset of size at most M .

Constraints

- $1 \leq n \leq 2 \times 10^5$
- $1 \leq M \leq n$
- $1 \leq B[i] \leq 10^9$
- All $B[i]$ are distinct
- $0 \leq G[i] \leq 10^9$

Example

Input

```
5 2
3 8 5 7 6
0 1 0 2 1
```

Explanation

The labels are $G[1] = 0$, $G[2] = 1$, $G[3] = 0$, $G[4] = 2$, $G[5] = 1$. Indices with the same label cannot both be selected (the subset would not be Valid). We select at most $M = 2$ indices from distinct label groups to maximise the Weight.

The optimal choice picks the heaviest representative from each label group: index 2 (label 1, value 8) and index 4 (label 2, value 7), giving Weight = 15.

Output

15

Sample Input	Output	Sample Input	Output
6 2 4 9 2 7 1 8 1 2 0 1 2 0	17	8 2 10 3 7 15 6 12 4 9 1 3 5 1 3 5 1 3	27
7 3 5 3 11 6 8 2 9 1 2 0 1 2 0 1	28	9 3 8 14 5 11 3 9 16 2 7 1 3 5 1 3 5 1 3 5	39

Problem B: Binary Search Game

Time limit : 1s
Memory limit : 256MB

Alice and Bob are playing a game. Their friend Charlie has chosen a number from $\{1, 2, \dots, n\}$. Alice and Bob will take turns guessing one of the numbers. If they are correct, they win. Otherwise, Charlie will tell them **higher** or **lower**.

Given n , what is the probability of Alice (first player) winning, if Alice and Bob play optimally?

Input

Input consists of only one positive integer n ($1 \leq n \leq 5000$) where Charlie has chosen a number from the set $\{1, 2, \dots, n\}$.

Output

A single floating (decimal) number which is within 10^{-6} of the actual answer.

Sample Input	Sample Output
1	1.0
2	0.5
3	0.66666666

Problem C: Contractor

Time limit : 1s
Memory limit : 256MB

Your company has been contracted to install street lights to light up a road of length n .

To maximize profit, you want to use up as many lights as possible. Lights can be installed anywhere on the road and light up 1 meter (diameter is 1, radius is 0.5) around it (if you install the light at the end of the road, half of the light will be wasted).

However, the government is not very supportive of your profit maximization efforts and hence has a rule that if you turn off any arbitrary street light, a certain section of the road (meter or less) will no longer be illuminated (ie, no street light can be completely redundant).

Input

Input consists of only of integer n .

Output

The maximum number of street lights we can install.

Sample Input	Sample Output
1	2
2	4

Problem D: Disrupted Energy System

Shinchan is experimenting with a peculiar energy system designed by Doraemon. The system starts at energy level 0 and evolves through a sequence of operations, never exceeding a maximum capacity M .

The system supports two types of **boost operations**, which can be applied any number of times:

- A *Type-I boost* increases the current energy by a fixed amount P ,
- A *Type-II boost* increases the current energy by a fixed amount Q .

Additionally, the system has two **disruption operations**, each of which can be used at most once during the entire process:

- A *Half Disruption*, which reduces the current energy to half (rounded down),
- A *Third Disruption*, which reduces the current energy to one-third (rounded down).

These disruption operations may be applied in any order, but each can be used at most once.

At every step, the energy must remain an integer in the range $[0, M]$.

Task

Let S denote the set of all energy levels that can be obtained starting from 0 by applying any valid sequence of operations.

Determine how many integers in the range $[0, M]$ are **not** contained in S .

Input

A single line containing three integers:

$$M, P, Q$$

Output

Output a single integer, the number of unreachable energy levels in $[0, M]$.

Constraints

- $1 \leq M \leq 5 \times 10^6$
- $1 \leq P, Q < M$

Example

Input

10 4 6

Output

0

Explanation

Combining all possibilities, the reachable set becomes: $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Sample Input	Sample Output	Sample Input	Sample Output
5 4 6	1	10 6 9	1
15 10 14	2	20 13 17	9

Problem E: Erina's Ramen

Time limit : 1s
Memory limit : 256MB

Erina has n ramen recipes. The i -th recipe uses broth $A[i]$ as the base. Note: Some ramen recipes may use the same broth.

She is setting up a stall at **CWI**'s annual fest. As the broth takes lot of time to cook and she has only 1 pot, all the ramen recipes on offer at a particular day must use the same broth.

As the kids at CWI are very picky, all the days will have to have pairwise distinct broths.

She may not be able to use all the recipes. It is possible that there are no days for some broths.

Finally, the number of recipes on the menu on each day is exactly twice as many as on the previous day and the first day can have arbitrary number of recipes on the menu. This is perhaps as the spoiled CWI kids demand increasing amount of variety.

Your task is to calculate the maximum number of recipes Erina could use throughout the days.

Note, that you should not maximize the number of days she can offer.

Input

The first line of the input contains one integer n ($1 \leq n \leq 10^6$) the number of recipes Erina knows.

The second line of the input contains n integers $A[1], A[2], \dots, A[n]$ where $A[i]$ is the base broth of the i -th ramen bowl.

Output

Print one integer, the maximum number of recipes that Erina could try out.

Sample Input	Sample Output
18 2 1 2 10 2 10 10 2 2 1 10 10 10 10 1 1 10 10	14
10 6 6 6 3 6 1000000000 3 3 6 6	9
3 1337 1337 1337	3

Problem F: Faulty Beacons

A drone is flying somewhere along an infinite straight runway, but its exact coordinate is unknown.

There are M beacons installed along the runway. Each beacon reports a constraint about the drone's coordinate, but some beacons may be malfunctioning.

Each beacon gives one of the following two types of readings:

- LEFT q : the drone is at a coordinate *at most* q ,
- RIGHT q : the drone is at a coordinate *at least* q .

Due to possible failures, not all beacon readings are guaranteed to be correct.

Task

Determine the minimum number of malfunctioning beacons.

Input

The first line contains a single integer:

$$M$$

Each of the next M lines contains:

$$s_j \ q_j$$

where:

- $s_j \in \{\text{LEFT}, \text{RIGHT}\}$,
- $0 \leq q_j \leq 10^9$.

Output

Print a single integer, the minimum number of beacons whose readings must be incorrect.

Constraints

- $1 \leq M \leq 1000$

Example

Input

```
3
RIGHT 2
LEFT 6
LEFT 3
```

Explanation

If the drone is at coordinate $y = 3$, all three constraints are satisfied.

Output

```
0
```

Sample Input	Sample Output	Sample Input	Sample Output
3 RIGHT 6 LEFT 2 LEFT 3	1	4 RIGHT 9 RIGHT 7 LEFT 2 LEFT 4	2
4 RIGHT 2 RIGHT 5 LEFT 9 LEFT 12	0	6 LEFT 3 LEFT 5 LEFT 7 RIGHT 2 RIGHT 4 RIGHT 6	2
2 RIGHT 7 LEFT 4	1	1 LEFT 1000000000	0
4 RIGHT 5 RIGHT 5 LEFT 5 LEFT 5	0	3 RIGHT 100 RIGHT 200 RIGHT 300	0